STAFF REVIEW · H446 261 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. RAM.

2. Holding the address of memory being accessed.

3. The position of an element within an array.

4. a[2]=3; a[1]=11; array [5,11,3,7], sum 26.

I Do — source worked steps

1. Predict — Immediate is literal 6; direct gets 4 from memory address 30.

2. Explain — Indirect follows pointer 30 stored at 31 to value 4; indexed adds offset 2 to base 30 to read 9.

We Do — guided CHECK

1. Predict
Immediate returns 8; direct reads M[40]=12.
Explain
M[41] points to address 42, so indirect reads 5. Indexed base 40+2 reads M[42]=5.

You Do — source mark schemes

1. [4] Immediate literal value in instruction [1]; direct location holds data [1]; indirect location holds another address [1]; indexed effective address is base+index [1].

2. [6] Immediate=4 [1]; direct M[10]=3 [1]; indirect first reads M[11]=10 [1], then M[10]=3 [1]; indexed address 10+2=12 [1], contents 9 [1].

3. [6] Array index 3 is fourth element [1]; each 16-bit element occupies 2 bytes [1]; displacement 3×2=6 bytes [2]; effective address 106 [1]; base+3 ignores element size [1].

Exit Ticket — source answer

1. False. Indirect addressing uses the pointer to find the actual operand location.